IDS 702 Unit 1 Quiz

Fall 2025

First name:	______________________	Last name:	______________________
Net ID:	______________________		

You are not permitted to use any materials (class notes, internet, etc.) on this assessment. Use of unauthorized materials or conferring with classmates will result in a 0 on the quiz and a report to the Office of Student Conduct.

> I hereby state that I have not communicated with or gained information in any way from my classmates or unauthorized materials during this assessment, and that all work is my own.
>
> Signature: ____

Multiple Choice

Circle the letter to mark your answer choice. Each question has one correct answer.

1. Which of the following scenarios is best modeled by the binomial distribution?

a. Jane has 3 board games and 6 card games on her shelf. She chooses 4 games without replacement. We want to know the probability that exactly 2 of them are board games.

b. The probability that Jane wins a single Solitaire game is 0.65. We want to know the probability that it will take 5 games for her to win 3 times.

c. The probability that Jane wins a game of Scrabble is 0.4, the probability that John wins is 0.3, and the probability that Jim wins is 0.3. We want to know the probability that, out of 5 games, Jane will win 2, John will win 2, and Jim will win 1.

d. The probability that Jane wins a single chess game is 0.65. We want to know the probability that she will win all 5 games in a competition.

2. Which of the following is **not** a step in creating a bootstrap sampling distribution?

a. Specifying the sampling distribution according to the central limit theorem

b. Sampling with replacement from the data

c. Calculating a sample mean for each bootstrap sample

d. Identifying the sample statistic that corresponds to the research question

3. Increasing the sample size __________________ the width of a confidence interval and increasing the standard error ____________________ the width of a confidence interval.

a. increases; increases

b. increases; decreases

c. decreases; increases

d. decreases; decreases

4. What is the distinction between standard deviation and standard error?

a. Standard error is the standard deviation of the sampling distribution

b. Standard deviation is the standard error of the sampling distribution

c. Standard error is the standard deviation of the population distribution

d. Standard deviation is the standard error of the population distribution

5. A p-value is the ________________ of observing ________________ as or more extreme than the observed, assuming the ________________ is true.

a. statistic; point estimate; alternative hypothesis
b. statistic; hypotheses; null hypothesis
c. probability; data; alternative hypothesis
d. probability; data; null hypothesis

6. Which of the following is guaranteed to be contained in a theoretical confidence interval of a mean?

a. The sample mean
b. The population mean
c. The critical value
d. The null value

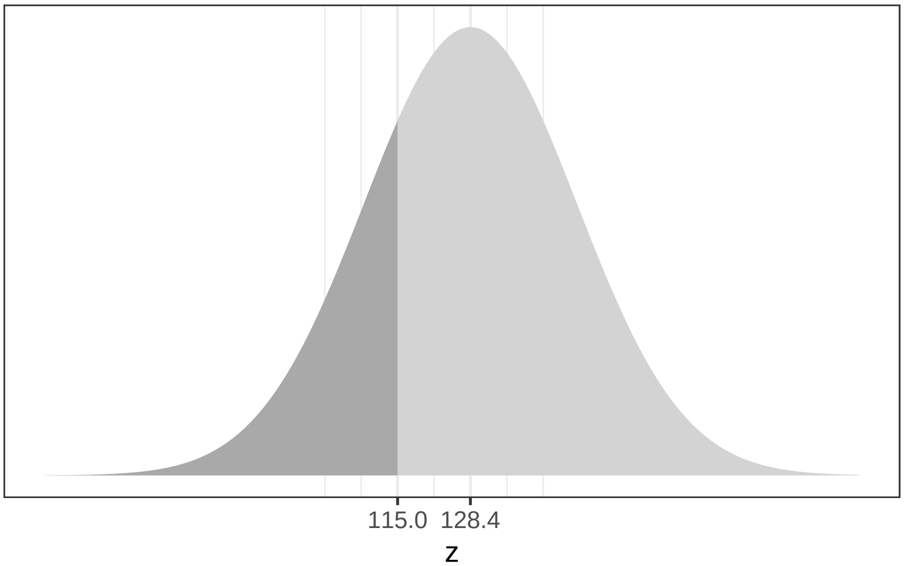
7. According to the central limit theorem, what is the relationship between the sample size and the standard error?

a. The standard error will increase as the sample size increases.
b. The standard error will decrease as the sample size increases.
c. The standard error will remain constant as the sample size increases.
d. The standard error is not related to the sample size.

8. What is the distinction between the joint distribution and the likelihood function?

a. The joint distribution describes the population, while the likelihood function describes the sample.
b. The joint distribution is obtained by maximizing the likelihood function.
c. The joint distribution is a function of the data, while the likelihood function is a function of the unknown parameter.
d. The joint distribution depends only on the parameter, while the likelihood function depends only on the data.

9. Systolic blood pressure of adults in the U.S. is normally distributed with mean 128.4 and standard deviation 19.6. If the probability that a randomly selected adult has blood pressure less than 115 is 0.25 (dark region below), what is the probability that a randomly selected adult has blood pressure greater than 115?



a. 0.75

b. 0.5

c. 0.25

d. There is not enough information to calculate the probability

True or False (circle one)

10. If two events are independent, their probabilities may not sum to 1.

True False

11. Sensitivity and specificity are examples of conditional probabilities.

True False

12. The log-likelihood function is often used instead of the likelihood function because it is easier to work with mathematically.

True False